LAB 16

# Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading

| Item | Detail |
|---|---|
| Topic | Topic 4: Building an End-to-End Algorithmic Trading Pipeline: Backtest, Forward Test and Performance Analysis |
| Pipeline stage | Automation |
| Duration | 30 min |
| Simulator page | http://localhost:8000/#/trading?new=1&strategy=ema_cross |
| Mode | No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only. |

*Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.*

## Scenario

The desk is ready to automate the configured EMA 10/30 strategy on SPY. Before any capital is considered, the pipeline must run unattended as a forward test: one decision cycle at a fixed time, completed bars only, pre-trade risk checks, a kill switch, idempotent paper orders and an audit row for every decision - first as a 2025 replay, then live on the Alpaca paper account during class.

## Objective

Create a bot, run it on a schedule, watch orders, positions and P&L live, and halt it with the kill switch.

## Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 16. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - you will use it in the debrief and in the later labs.

## Lab Steps - In the Trading Simulator

*No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 16: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.*

### Step 1. Create the bot

Paper Trading > + New bot: EMA crossover 10/30 on SPY, schedule 'every 10s', simulator broker, replay from 2025-01-02.

Open: http://localhost:8000/#/trading?new=1&strategy=ema_cross (Paper Trading > New bot (EMA cross))

*Check: The bot appears with status stopped.*

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### Step 2. Start it and watch

Press Start. Every cycle replays one trading day: it sees bars to yesterday's close and its orders fill at today's open. Or press 'Run one cycle now' repeatedly. (If the bot does not cycle, the simulator was started without its scheduler - ask your trainer.)

Open: http://localhost:8000/#/trading

*Check: Equity curve grows cycle by cycle; the Pipeline Map path lights up during each cycle.*

### Step 3. Kill switch

Press the bot's Kill switch, then try 'Run one cycle now'. Release it. Try the global kill switch in the top bar too.

Open: http://localhost:8000/#/trading

*Check: Cycles report 'kill switch ON - no orders sent'; the Risk stage turns red on the Pipeline Map.*

### Step 4. Backtest vs forward

Press Compare in 'Backtest vs forward test'.

Open: http://localhost:8000/#/trading

*Check: The curves track closely; you can explain the remaining gap (whole shares, fills, timing).*

**Expected in the app: A bot equity curve, a blotter of its orders, a backtest-vs-forward comparison and a tested kill switch.**

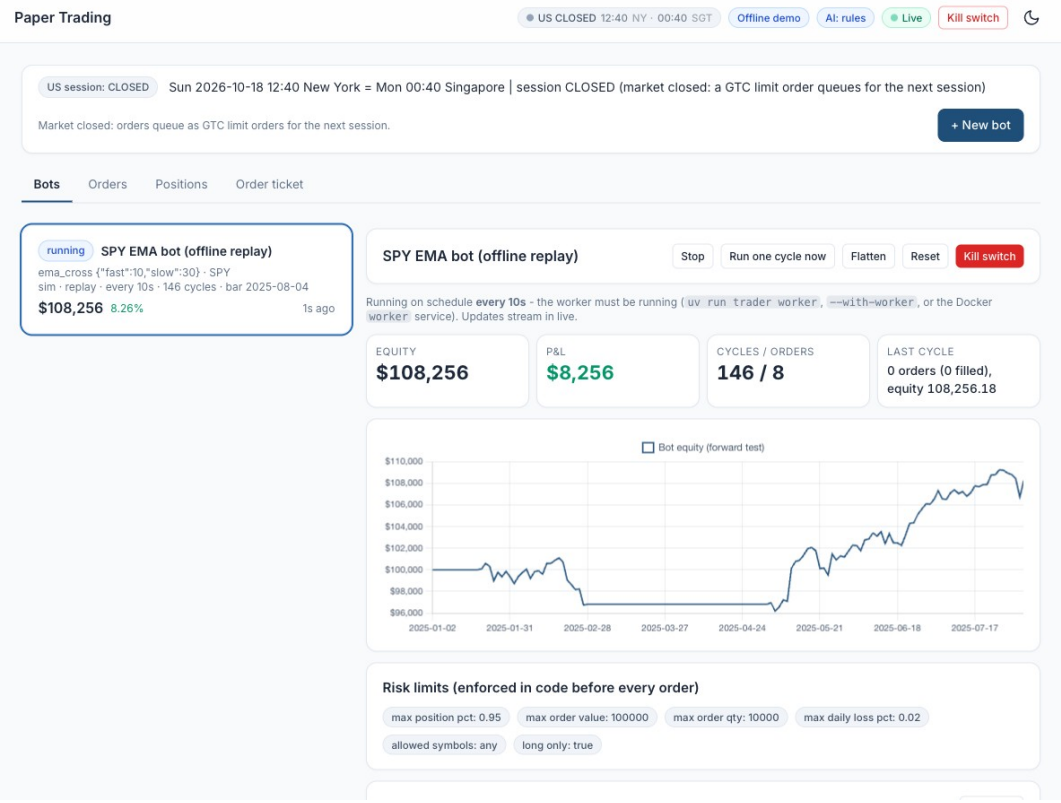



*Figure: Lab 16 in the Trading Simulator.*

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## Optional: Let AI Write the Code

*This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs/Prompt_Library.html and .pdf.*

| A risk-checked paper-trading bot - run it in Google Colab |
| --- |
| I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: write a simple paper-trading bot for SPY: each run it takes daily bars up to yesterday, computes an EMA 10/30 crossover, checks risk limits (maximum position 50% of equity, no new buys after a 2% daily loss, a KILL_SWITCH flag) and changes the position only when the target changes, logging every decision to a CSV. Include a replay mode that simulates 2025 day by day with a $100,000 paper account (no broker needed) and compares the forward result with a backtest of the same rule. |

Open in Claude | Open in ChatGPT | Open Google Colab

## Guide: Paper Trading on Alpaca, Step by Step - Manual and Automated

First place one order by hand to see the order lifecycle, then let the simulator place orders automatically (Labs 8 and 16).

### 1. Place a manual paper order

In the Alpaca dashboard select the Paper Trading account (upper left), search the symbol (for example SPY) and open its page. Set the side, quantity, order type and time in force, then press Buy.

### 2. Follow the order lifecycle

Open the Orders view and watch the status move from new / accepted to filled (or cancelled). Open Positions to see the holding, cost basis and unrealised P&L.

### 3. Know the trading hours from Singapore

The US regular session runs 21:30-04:00 Singapore time (22:30-05:00 in the northern winter). Outside it, Alpaca's 24/5 trading accepts only LIMIT orders with extended hours enabled (time in force DAY or GTC), at lower liquidity; a market order placed while the market is closed waits for the next session.

### 4. Let the simulator place the orders

In the simulator open Paper Trading > Order ticket (Lab 8): choose the symbol, side, quantity and order type and press Send. With Alpaca connected it becomes a real paper order on your Alpaca account.

### 5. Run the strategy automatically

Paper Trading > + New bot (Lab 16): pick a strategy and symbol, set the schedule and press Start. Each cycle fetches bars, computes the signal, runs the risk checks and places or skips the paper order.

### 6. Check the results in the dashboard

Refresh the Alpaca dashboard: the bot's orders appear under Orders and its holdings under Positions, matching the simulator's Orders tab and the bot's trade log.

### 7. Stop it safely

Press the bot's Kill switch (or the global kill switch in the simulator's top bar): the next cycle halts before any order is sent. Cancel working orders and close positions if you wish.

*From Singapore the US regular session is 21:30-04:00 (22:30-05:00 in northern winter); class-time orders use Alpaca's 24/5 extended-hours limit orders.*

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## Debrief

The runner is a software agent coded from a selected model: signal on completed bars, deterministic risk limits, idempotent execution, a kill switch and an audit log. The forward test and the live paper loop check the actions it takes and their outcomes on unseen data, including the real effect of whole shares, fills and session rules.

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